FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-064C - CampusSpark Spark Gate

SEALED fresh full retest - explicit assignment only.

Artifact	C3-064C
Status	2026 - LEARNER EDITION
Phase	C3-05 Distributed Processing with Spark | 16 lessons | 46-60 focused hours
Prerequisite	C3-04 Gate PASS; C2B SQL/Python/modeling foundations assumed, distributed execution taught here.
Current lane	Apache Spark / PySpark 4.2.0 | Python 3.10+ | Java 17+ | learner local[2]
Brand	Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6 | AMOLED #000

House teaching rule: mental picture -> tiny worked example -> guided real Spark workflow -> expected result -> why it worked -> break/fix -> independent changed task -> validation -> physical-plan/production reasoning -> evidence. A green Spark job is not mastery if grain, quality, plan or reproducibility is wrong.

CampusSpark scenario

Meal transactions join dining-hall metadata. One price is missing and H5 is intentionally hot. This is a fresh full retest with different identifiers, metrics and business semantics.

Gate B/C remains sealed after a failed prior Gate and after remediation. It opens only when Mentor/reviewer explicitly assigns a fresh full retest.

Independent requirements

• Use explicit schemas and the laptop-safe local Spark profile.

• Create clean + rejected/flagged outputs; never silently delete suspect data.

• Prove dimension join-key uniqueness before joining.

• Produce grouped operational metrics at a stated output grain.

• Add one deterministic window metric that preserves fact-row grain.

• Write curated Parquet and justify partition/file strategy.

• Inspect a formatted plan and identify shuffle/join operators.

• Measure key distribution and classify hot-key/extreme behavior as skew risk, data-quality risk, or both.

• Demonstrate one evidence-based performance decision involving broadcast, repartition/coalesce, caching or AQE.

• Provide at least two automated transformation/invariant tests.

• Package a repeatable CLI-style job with version/run/output validation evidence.

• Explain one design change when moving from local[2] to a real cluster/Fabric Spark runtime.

Competency Gate - every competency required

Competency	PASS evidence
Correctness & requirements	Observed, validated, reproducible and explainable on this fresh case.
Data integrity & validation	Observed, validated, reproducible and explainable on this fresh case.
Reliability & rerun safety	Observed, validated, reproducible and explainable on this fresh case.
Testing & change safety	Observed, validated, reproducible and explainable on this fresh case.
Observability & diagnosis	Observed, validated, reproducible and explainable on this fresh case.
Security & evidence hygiene	Observed, validated, reproducible and explainable on this fresh case.
Maintainability & reproducibility	Observed, validated, reproducible and explainable on this fresh case.
Design reasoning & ownership	Observed, validated, reproducible and explainable on this fresh case.

PASS only if every required competency passes AND zero unresolved critical failures remain.

Evidence packet

Evidence	Your saved proof/path/note
Spark/Python/Java/session version proof	
Input/output grain statement	
Explicit schema evidence	
Clean/rejected reconciliation	
Dimension-key uniqueness + join row-count proof	
Grouped metric validation	
Window metric grain/tie rule	
Formatted plan annotation	
Skew/key-distribution evidence	
One controlled performance decision	
PySpark tests/invariants	
Parquet output/file evidence	
CLI/runbook/handoff	
Cluster/Fabric migration reasoning	